

SE Practices 1 | SEP1

Summary

CONTENTS

- 1. Object-Oriented Analysis and Design 2
- 2. Unified Modeling Language (UML) 2
- 3. Unified Process (UP) 2
- 4. Iterative Development 3
- 5. Domain Models 3
- 6. Requirements 3
 - 6.1. Use Cases 3
 - 6.1.1. Actors 3
 - 6.1.2. Formats 4
 - 6.1.3. Elementary Business Processes (EBP) 4
 - 6.1.4. Finding Use Cases 4
 - 6.1.5. System boundary 4

1. OBJECT-ORIENTED ANALYSIS AND DESIGN

“Responsibility-driven design”

- 1) Requirements analysis: What should the product do?
 - Software Requirements Specification (SRS)
- 2) Domain analysis: What is the problem domain?
 - Static View: Domain Model (UML notation: Class Diagram)
 - Dynamic View: Behavioural Models (UML notation: Interaction Diagram, State Machine Diagram, Activity Diagram, etc. depending on the domain)

TODO:

diagrams oo slides 6

TODO:

oo slides in general

Analysis: Investigation of the objects in the problem domain

Design: Defining software objects and how they collaborate to fulfill the requirements

- 1) Define Use Cases
- 2) Define a Domain Model
- 3) Assign Object Responsibilities and Draw Interaction Diagrams
- 4) Define Design Class Diagrams

2. UNIFIED MODELING LANGUAGE (UML)

Visual language for specifying, constructing and documenting the artifacts of systems.

Perspectives to apply UML (from abstract to specific): Conceptual, Specification, Implementation

UML Notation for Domain Models:

- Classes for sets of similar objects in the problem domain
- Attributes for properties of those objects
- Associations for relationships between those objects
- Generalisations in between classes with a “each X is a Y” relationship

3. UNIFIED PROCESS (UP)

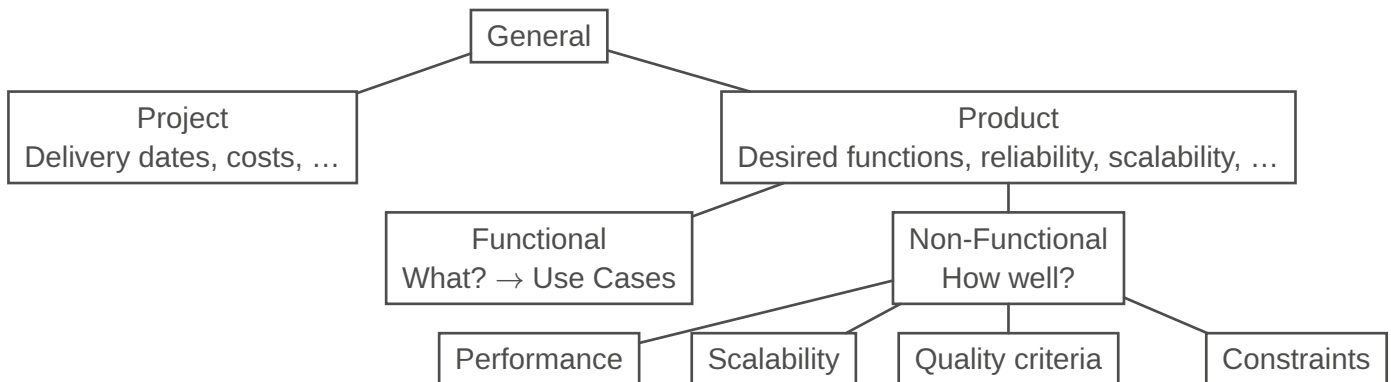
An iterative software development process for building object-oriented systems.

Software development process: describes an approach to building, deploying and possibly maintaining software.

4. ITERATIVE DEVELOPMENT

5. DOMAIN MODELS

6. REQUIREMENTS



Discipline: a systematic approach to finding, documenting, organizing, and tracking the changing requirements of a system

Work Products: Software Requirement Specification (SRS) = Capabilities/Conditions that the software must fulfil

6.1. USE CASES

“A set of use-case instances, where each instance is a sequence of actions a system performs that yields an observable result of value to a particular actor.”

- A textual process description with a concrete purpose
- Documents a story of some actor using a system to meet goals
- Places requirements within the context of a user with goals.
- Simple & familiar for the average stakeholder

UseCaseName

A use case is typically composed of:

- A main success scenario, and
- Multiple extensions / alternative scenarios

6.1.1. Actors

An actor is something with behaviour, such as a person (identified by a role), computer system, or organization.

Actors are associated to roles, and not to physical entities

ActorName

<<actor>>
ExternalSystem
Name

Primary Actors: Actors whose goals are fulfilled through using services of the SuD.
E.g., cashier in “Process Sale”

Supporting Actors: Provides a service to the SuD
E.g., Credit card billing system in “Process Sale”

Offstage Actors: Has an interest in the behaviour of the use case, but is not primary or supporting.
E.g., Government tax agency in “Process Sale”

6.1.2. Formats

Brief: Short, one-paragraph summary, usually of the main success scenario

Casual: Multiple paragraphs that cover various scenarios

Fully dressed: All scenarios, steps and variations written in detail, in a specific form, with supporting sections, such as preconditions and success guarantees.

TODO:

W3 S12

6.1.3. Elementary Business Processes (EBP)

Use Cases should be identified at the EBP level:

A task performed by one person in one place at one time, (single session) in response to a business event, which adds measurable business value (Boss Test) and leaves the data in a consistent state

6.1.4. Finding Use Cases

- 1) Fix the system boundary: Software, HW/SW-System, Entire Organisation?
- 2) Identify primary actors and their goals: At the EBP level
- 3) Write down the use cases – first in the brief Format

6.1.5. System boundary

TODO:

W3 S19+